



Cameras now size of a grain

Researchers create an ultracompact camera the size of a coarse grain of salt.
<https://digit.in/dec21-22>



Solar 'Farm' Farm

This Colorado 'solar garden' is literally a farm under solar panels.
<https://digit.in/dec21-23>



Origins of Heatsinks

Delve into the history of heatsinks

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INTRODUCTION

You probably read the title and are thinking, why learn about the history of heatsinks? Well, the story is more interesting that you may think.

There is technology everywhere these days and all that technology generates heat. In fact, anything with a processor generates heat and that heat needs to be dissipated in an efficient manner so that the appliance keeps working as expected. This is where heatsinks come in. For the uninitiated, the picture on the right shows how one usually looks. Now that you know what it looks like, let's start learning about the history of heatsinks.

THE HISTORY

The processors of the yesteryears didn't generate enough heat for them to require a heat sink. The heat simply dissipated through the printed circuit board by taking advantage of the CPU package made of plastics, through the socket. Later, as CPUs upgraded in integration scale and the design rule for circuit patterns increased along with clock frequency, the heat generation continued to surge and soon, the world



An example of a Die Cast Heatsink

felt the need for an external solution to dissipate heat and that is when heat-sinks based on extrusions and die castings were employed for desktop PCs.

Then, since us humans are never satisfied, we called for upgrades, and with upgrades, you'll never guess what happened. MORE HEAT!

More heat meant we needed better cooling, that brought us to including a motor driven fan into the heatsink to force airflow through it so it cools down faster. We are still using this method with tons of improvements and even though water cooling has been introduced, we're still not at the point of using only water cooling. Air cooling is still the staple in PCs and other electrical appliances.

On the other hand, in laptops, heat was diffused over the entire casing of it through an aluminum sheet for it to be naturally dissipated into the air around it. Then, in 1993, a metal pipe was included in the ideal heat dissipation path which helped cool things down way better. Yet again, there were upgrades, so when laptops started going boom boom due to heat in testing, motor driven fans were included.

It worked like a charm and it is still used as the default method for cooling laptops to this day with a few improvements, of course.

Now, desktop CPUs generate a heat rate of over 100W, and the CPUs made specifically for laptops, for example the special H series chips from Intel, generate a heat rate of over 30W. And of course, how can we forget the technology that is rarer than a blue moon, the GPU! Their heat generation rate is over 10W which tends to heat things up in the casings they are held in.

I'm sure you get the point, heatsinks are basically used in tons of crucial appliances from which extremely high performance is expected.

NEW REQUIREMENTS FOR HEATSINKS

The heatsinks we have today are required to have high thermal performance as well as other advantageous characteristics. For